Alfredo Pasten Jr

pastenalfredojr@gmail.com | linkedin.com/in/alfredo-pasten-jr | shadowalzazel.dev

Experience

Teaching Assistant (Instructional Student Assistant) | *CSULB – Python, SQLite* Aug. 2025 – Jan. 2026
- Deployed a **SQLite**-backed tool that syncs **Zybooks** grades to Canvas, eliminating manual data entry for faculty.
- Collaborated with faculty on course logistics and independently led proctoring responsibilities.

Projects

Invictus: Astrum | *Python, FastAPI, NumPy, Numba, Svelte, Three.js* Ongoing
- Physics-driven **RTS** on a true-scale star-system simulation (Keplerian orbits, JPL DE421 ephemerides) with station keeping, refuel orders, sensor detection, and command relay; **Numba** kernels run ∼100× faster than pure Python.
- Multiplayer lobbies with server-enforced fog of war, 20 Hz **WebSocket** pushes, and rate-limited orders; 243 tests.

Null Colony | *Python, Svelte, MCP, JSON-RPC, Caddy* | nullcolony.net Ongoing
- Public lobby for existing AI agents and people: each self-hosted hotel exposes one **MCP** endpoint (JSON-RPC over HTTPS) with tools to list floors, enter key-locked rooms, chat, whisper, pin notes, and hand off.
- Restart-proof guest sessions, stable name-based addresses, idle-seat release, and browser seats; stdlib Python.

Multi-modal Agent Assistant Himiko | *Ollama, ElevenLabs, PyTorch, Whisper, Discord* Ongoing
- Architected a multi-modal AI agent for Discord, VTube Studio, and Minecraft with real-time **Whisper** voice input and a custom **StyleTTS2** voice.
- Designed a plugin system so modules (e.g. **DeepL** translation) are added without modifying core agent logic.

ML Cryptocurrency Analyzer | *PyTorch, Graph ML, Elliptic Dataset* May. 2026
- Led a graph ML project classifying illicit Bitcoin transactions on the Elliptic dataset with **PyTorch** GNNs.

Server-side Minecraft Mod | *Kotlin, Java, PaperAPI, Blockbench* Ongoing
- Built a server-side **Minecraft** expansion (plugin, datapack, resource pack) adding items, biomes, dimensions, dungeons.
- Published the plugin, ran a 40+ member server community, and integrated the Dropbox API for asset delivery.
- Built **Docker**ized multi-agent bots (**JS-Pybridge**, **Ollama**); contributor to open-source MindCraft.

Leadership

Association for Computing Machinery – Design Lead Aug. 2023 – May. 2024
California State University – Long Beach *Long Beach, CA*
- Directed CS Department design work: a club info portal and Discord authentication for students and guests.
- Produced promotional materials and event branding for general body meetings and large-scale events.

Paul Merage Future Leaders Initiative May. 2023 – Aug. 2023
University of California – Irvine *Irvine, CA*
- Selected for an emerging-tech leadership program; presented to the Mayor of Irvine and an NVIDIA VP.
- Facilitated leadership workshops and built a cross-institutional network of peers and mentors.

Education

California State University – Long Beach Long Beach, CA
Master of Science in Computer Science *Jan. 2027 – Expected*
Bachelor of Science in Computer Science, AI/ML Track *Aug. 2023 – Dec. 2026*

Hanyang University Seoul, South Korea
Hanyang International Winter School *Dec. 2025 – Jan. 2026*

Long Beach City College Long Beach, CA
Associate Studies in Computer Science *Aug. 2021 – Jun. 2023*

University of California – Berkeley Berkeley, CA
Undergraduate Studies in Electrical Engineering and Computer Science *Aug. 2020 – May. 2021*

Technical Skills

Languages: Python, Kotlin, TypeScript, JavaScript, Java, C#, C++
AI/ML: PyTorch (GNNs), Whisper, StyleTTS2, ElevenLabs, Ollama, OpenAI API, MCP tool servers (JSON-RPC)
Simulation & Scientific: NumPy, Numba JIT, Skyfield + JPL DE421, JPL Horizons API, Astropy, pandas, Parquet
Backend & Infra: FastAPI, WebSockets, Uvicorn, Caddy, Docker Compose, SQLite, pytest, Gradle, PaperMC API
Frontend: Svelte 5, SvelteKit, Three.js, Vite, React, Expo; deployed on Vercel and Netlify